Convolutional Neural Networks with Keras: An Experimental Study for MNIST and CIFAR-10 Image Classification

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***Abstract*—This paper presents a reproducible experimental study of convolutional neural networks (CNNs) implemented with Keras for image classification. The study first compares a multilayer perceptron (MLP) with a basic CNN on the MNIST dataset, then evaluates a deeper CNN architecture on CIFAR-10. The experiments examine training behavior, confusion matrices, optimizer selection, regularization strategies, training data volume, and visual interpretability through learned filters, feature maps, and Grad-CAM. The final CNN achieved 99.27% accuracy on MNIST and 81.90% accuracy on CIFAR-10. A data-scaling experiment showed that CIFAR-10 accuracy increased from 46.69% using 10% of the training data to 70.98% using the full training set. An exploratory hybrid CNN-attention model was also evaluated; however, under the current configuration, it performed below the final CNN. Overall, the results show that CNNs can outperform an MLP on MNIST and achieve competitive performance on CIFAR-10, while emphasizing the importance of systematic ablation, sufficient training data, and careful interpretation of visualization-based analyses.**

***Index Terms*—Convolutional Neural Networks, Keras, MNIST, CIFAR-10, Image Classification, Grad-CAM, Feature Maps, Data Augmentation, Deep Learning.**

I. Introduction

In computer vision, image classification is a typical benchmark task [1]. The aim is to assign an input image to a single class from a predetermined set. Although it flattens pictures early and does not directly maintain spatial architecture, a thick neural network is capable of classifying images. Because CNNs handle local areas and learn feature representations over several layers, they are more suitable for image data [1].

In this study, Keras [2] was used to create a repeatable student experimental method to investigate CNNs. The objectives are not to propose a novel architecture or assert cutting-edge performance. Rather, my goal is to comprehend how fundamental model and training decisions impact MNIST and CIFAR-10 performance [1], [3]. To compare an MLP baseline and a CNN in a more straightforward scenario, I began with MNIST. Subsequently, I go on to CIFAR-10, which is more difficult owing to its tiny color pictures, more item appearance variance, background clutter, and visually comparable classes.

The primary contributions of this study are as follows:

- An analysis of optimizer selection, ablation experiments, and training data quantity
- A visual interpretation study using learned filters, feature maps, and Grad-CAM.
- An exploratory CNN-attention model and parameter-level comparison with VGG16, VGG19, and Mo-bileNetV2.
- A comparison between an MLP baseline and CNN on MNIST.

The remainder of this paper is structured as follows. The background most pertinent to this research is briefly reviewed in Section II. The datasets, preparation procedures, models, and assessment measures are described in Section III. The findings of this study are presented in Section IV. Visual interpretability is the main topic of Section V. The key findings, constraints, and potential expansions are discussed in the concluding sections.

II. Related Work

AlexNet demonstrated the efficacy of deep CNNs for large-scale image categorization [4]. Deeper networks constructed from repeated tiny convolutions might also perform well, as shown later by VGG16 and VGG19 [5]. By using depth-wise separable convolutions and inverted residual blocks, MobileNetV2 focuses more on efficiency [6]. Grad-CAM is a popular visualization technique that highlights the areas of an image that contribute to a prediction [7]. The primary context for the architectures and interpretation tools used in this investigation was provided by these studies.

III. Methodology

A. Datasets

The experiments used two standard datasets: MNIST and CIFAR-10 [1], [3]. MNIST comprises grayscale photographs of handwritten numbers, each measuring 28×28 pixels, categorized into 10 classes. CIFAR-10 comprises color pictures measuring $32 \times 32 \times 3$ across ten categories: aircraft, car, bird, cat, deer, dog, frog, horse, ship, and trucks.

These datasets were selected because they provide two distinct levels of complexity. MNIST is crucial for verifying

the proper functionality of the fundamental pipeline. CIFAR-10 presents more difficulty because of the increased variability in color, object appearance, and backdrop within the pictures.

Figures 1 and 2 show examples from the MNIST and CIFAR-10 datasets, respectively.

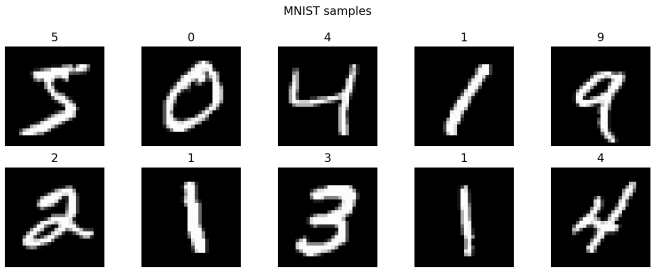

Fig. 1. Examples from MNIST dataset.

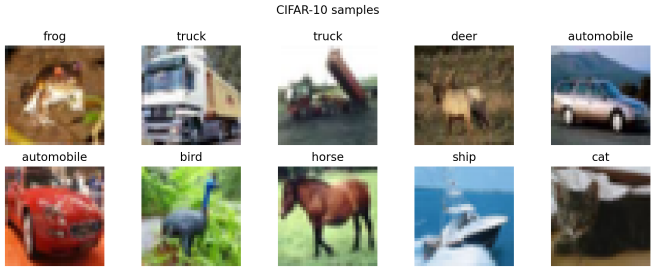

Fig. 2. Illustrations from CIFAR-10 dataset.

B. Data Preprocessing

All photos were normalized to the range $[0, 1]$. The MNIST images were altered to incorporate a single channel. The CIFAR-10 photos were maintained in the RGB format. For the final CIFAR-10 CNN, I additionally performed data augmentation using random horizontal flips, minor rotations, zooming, and translations. The purpose was to evaluate whether tiny alterations in the training pictures may assist generalization.

C. Models

Four model types were utilized in this study: an MLP baseline, a basic CNN for MNIST, a deeper CNN for CIFAR-10, and an experimental hybrid CNN-attention model.

1) MLP Baseline: The MLP baseline first flattens the input image, then applies dense layers followed by a softmax classifier. This model was used as a basic reference on MNIST to compare a fully connected approach with a convolution-based one.

2) MNIST CNN: The MNIST CNN employs two convolutional blocks with ReLU activations and max-pooling, followed by a dense classifier. This model is purposely simplistic and is designed to investigate if retaining local spatial structure provides a discernible benefit over the MLP baseline.

3) CIFAR-10 CNN: The final CIFAR-10 CNN used three convolutional blocks with 32, 64, and 128 filters. Each block has two convolutional layers with ReLU activations and max-pooling, and some experimental variations additionally incorporate batch normalization or dropout. The classifier concluded with a dense layer and softmax output. This model was the core architecture utilized for the CIFAR-10 tests.

4) Hybrid CNN-attention: The hybrid model initially collects local visual characteristics using convolutional layers. The generated feature maps are then molded into a sequence and provided to a multi-head attention layer [8]. This model was included as an experimental addition to determine whether increasing attention in this basic situation affects the classification performance.

D. Evaluation Metrics

The models were tested using accuracy, macro F1-score, weighted F1-score, training and validation curves, and confusion matrices. Accuracy is utilized as the key overall metric. The macro F1-score is provided to assess whether the performance is appropriately balanced across classes instead of being dominated by easier classes. Confusion matrices are useful for error analysis, notably on CIFAR-10, where some classes are visually identical.

IV. Experimental Results

This section reports the main results of this study. I first compared the MLP and CNN on the MNIST dataset. Then, I present the CIFAR-10 results, followed by an optimizer comparison, ablation experiments, training data scaling, and the exploratory hybrid model.

A. MNIST: MLP versus CNN

Table I compares the MLP and CNN on the MNIST dataset. In this experiment, the CNN performed better than the MLP in terms of accuracy and macro F1-score. This finding shows that convolutional layers are better suited to picture data because they maintain local spatial information.

TABLE I
MNIST Performance Comparison.

Model	Dataset	Accuracy	Macro F1
MLP	MNIST	98.09%	98.08%
CNN	MNIST	99.27%	99.26%

The training curves in Fig. 3 indicate that the MNIST CNN converged rapidly and that the validation accuracy remained close to 99% throughout the training. The confusion matrix in Fig. 4 is mostly concentrated on the diagonal, indicating that most test digits are classified correctly.

challenging: the images are small, the objects vary more, and several classes are visually similar to each other.

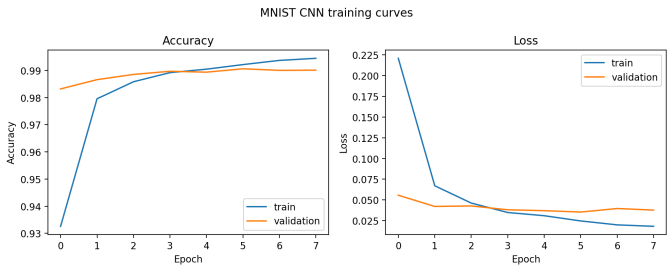

Fig. 3. Training and validation curves for MNIST CNN.

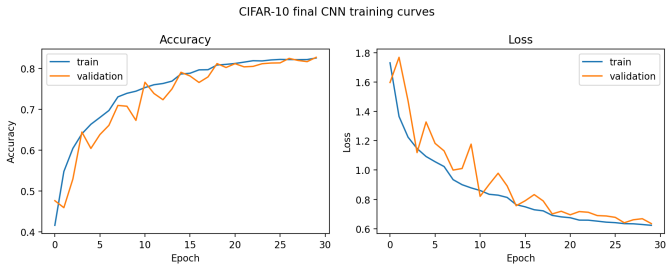

Fig. 4. Training and validation curves for the final CIFAR-10 CNN.

The confusion matrix in Fig. 6 demonstrates that many predictions fall on the diagonal, but noticeable off-diagonal errors remain. The most problematic examples occur among comparable animal types, such as cats, dogs, deer, horses, and birds. Some misunderstandings are also noticeable across vehicle-related classifications, such as aircraft, ships, cars, and trucks.

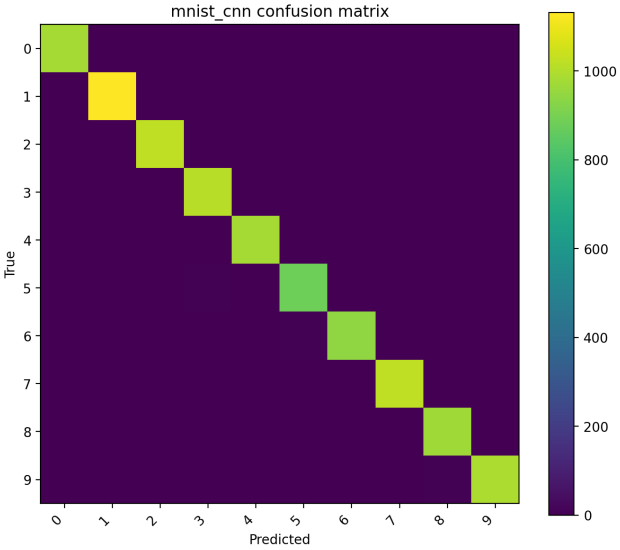

Fig. 5. Confusion matrix for MNIST CNN.

B. CIFAR-10 Final CNN

Table II presents the primary results for MNIST and CIFAR-10. The final CNN achieved an accuracy of 81.90% and a macro F1-score of 81.63 % %on CIFAR-10. The hybrid CNN-attention model obtained an accuracy of 67.07% and hence remained below the final CNN in the current setup. Therefore, the hybrid model is treated as an exploratory extension rather than an improvement over the main CNN.

TABLE II
MAIN EXPERIMENTAL RESULTS.

Model	Dataset	Accuracy	Macro F1
MLP	MNIST	98.09%	98.08%
CNN	MNIST	99.27%	99.26%
Final CNN	CIFAR-10	81.90%	81.63%
Hybrid CNN-Attention	CIFAR-10	67.07%	66.85%

Fig. 5 shows the training behavior of the final CNN on CIFAR-10. The training and validation accuracies both increased throughout training, whereas the loss decreased. Compared with MNIST, the validation curve is less stable, which is reasonable because CIFAR-10 is more

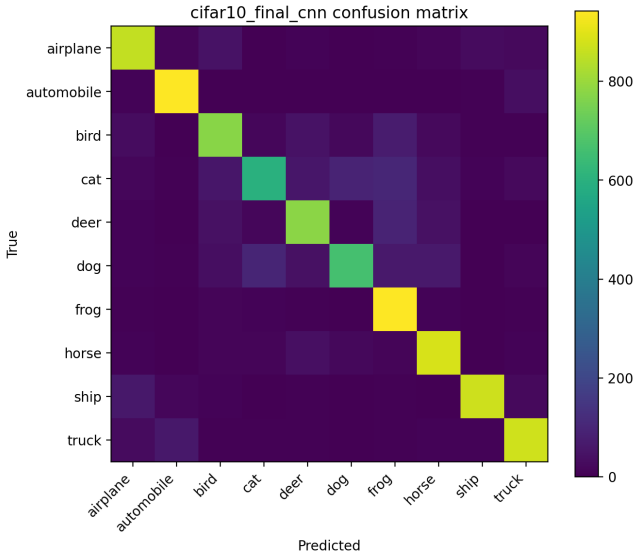

Fig. 6. Confusion matrix for the final CIFAR-10 CNN.

C. Optimizer Comparison

Table III compares Adam, AdamW, and SGD with momentum on the CIFAR-10 dataset. In this scenario, AdamW delivered the greatest accuracy, followed by Adam, whereas SGD performed worse under the measured hyperparameters. I do not consider this a general rating between optimizers. This merely indicates that, for this model and training schedule, the adaptive optimizers performed better than SGD with the given parameters.

TABLE III
OPTIMIZER COMPARISON ON CIFAR-10.

Optimizer	Learning Rate	Accuracy	Macro F1
Adam	0.001	71.71%	71.21%
AdamW	0.001	72.73%	71.99%
SGD + momentum	0.010	67.36%	65.88%

D. Ablation Study

Ablation experiments were conducted to examine the effects of the individual training components. Table IV demonstrates that dropout delivers the greatest single improvement in this set of experiments. Weight decay and batch normalization similarly enhanced the baseline, whereas the data augmentation variation was marginally lower than the baseline in this experiment. The entire model did not outperform the best individual variations. This finding is interesting because it indicates that combining different regularization techniques does not automatically yield optimal performance without additional adjustments.

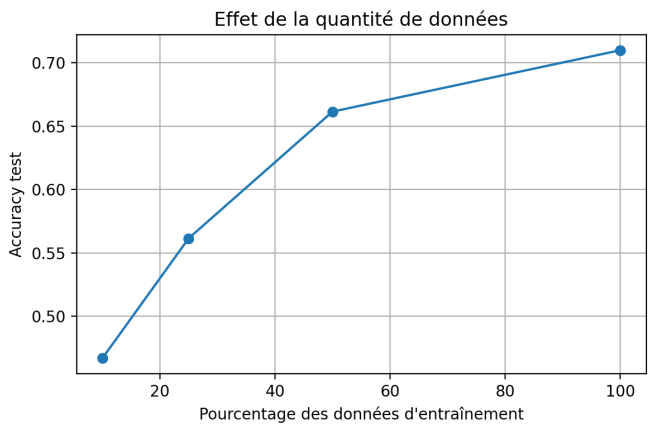

Fig. 7. Effect of training data quantity on CIFAR-10 accuracy.

E. Hybrid CNN-Attention Model

TABLE IV
ABLATION STUDY ON CIFAR-10.

Experiment	Accuracy	Macro F1
Baseline	71.11%	70.71%
Data augmentation	70.00%	69.03%
Dropout	75.81%	75.39%
Weight decay	73.71%	73.41%
Batch normalization	73.29%	73.37%
Full model	70.38%	69.14%

F. Effect of Training Data Quantity

The hybrid CNN-attention model was used for the experiments. Fig. 8 illustrates that the model improved throughout the training, but its ultimate performance remained below that of the final CNN. In the given setting, more attention does not result in a better classifier. A likely explanation is that this architecture requires more tuning and possibly a different training strategy before a fair comparison can be made.

To test the influence of dataset size, I trained the identical CIFAR-10 configuration using 10%, 25%, 50%, and 100% of the training data points. Table V and Fig. 7 indicate a noticeable improvement in accuracy when additional training data were employed. The increase was notably high between 10% and 50%, whereas the growth from 50% to 100% was still positive but more gradual.

TABLE V
EFFECT OF CIFAR-10 TRAINING DATA QUANTITY

Training Data	Samples	Accuracy	Macro F1
10%	5,000	46.69%	44.55%
25%	12,500	56.15%	54.68%
50%	25,000	66.14%	65.16%
100%	50,000	70.98%	70.75%

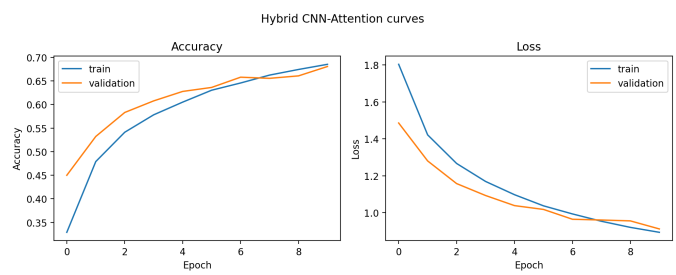

Fig. 8. Training and validation curves for the hybrid CNN-attention model.

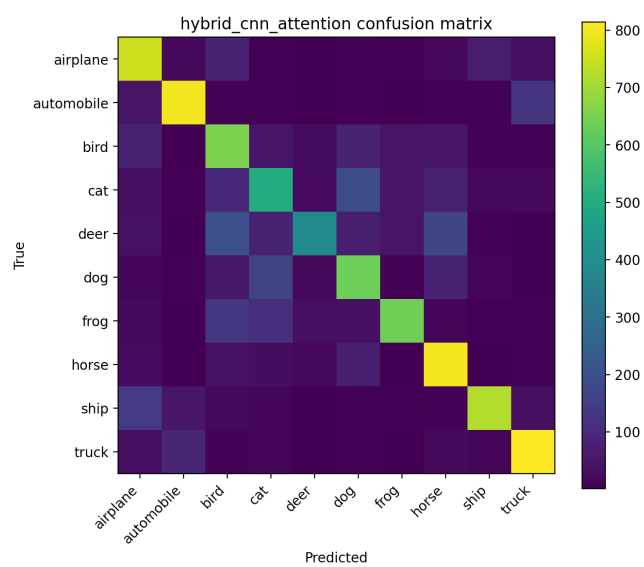

Fig. 9. Confusion matrix for the hybrid CNN-attention model.

G. Reference Architectures: VGG16, VGG19, and MobileNetV2

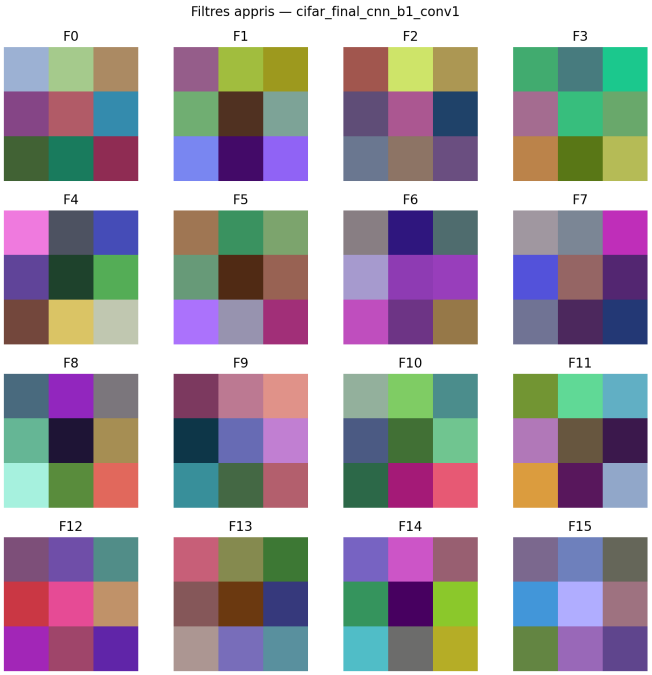

Fig. 10. The filters were learned from the first convolutional layer.

VGG16, VGG19, and MobileNetV2 were provided solely as reference architectures. In this study, I did not train them using pretrained ImageNet weights; hence, I did not utilize them for direct performance comparison. Instead, Table VI presents the suggested student-level models in a larger architectural context. The parameter counts show that VGG16 and VGG19 are much heavier than MobileNetV2, which is consistent with their different design objectives.

TABLE VI
PARAMETER-LEVEL COMPARISON OF REFERENCE ARCHITECTURES.

Model	Total Parameters	Role
VGG16	14.72M	Deep reference model
VGG19	20.03M	Deeper reference model
MobileNetV2	2.27M	Lightweight model

V. VISUAL INTERPRETABILITY

A. Learned Filters

Fig. 10 displays the learned filters from the first convolutional layer of the CIFAR-10 CNN. These filters are not immediately interpretable as object parts. Instead, they appear to respond to simple local patterns, such as color changes, edges, and texture-like variations, which is consistent with the usual role of early CNN layers.

B. Feature Maps

Fig. 11 shows the feature maps from an intermediate convolutional layer. The maps do not react to the same picture areas. Some activations accentuate edges or textured regions, whereas others highlight larger patterns. This visualization does not fully explain the model, but it provides a qualitative view of how different channels react to the input image.

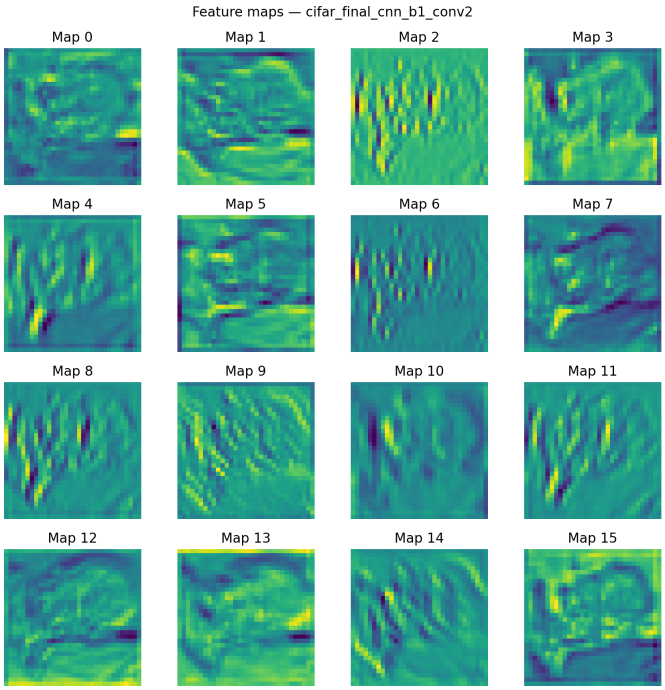

Fig. 11. Intermediate feature maps from CIFAR-10 CNN.

C. Grad-CAM Error Analysis

Grad-CAM was used to examine a misclassified CIFAR-10 case. In Fig. 12, the real class is dog, whereas the model predicts horse with a confidence of 0.65. In this situation, the heatmap is poor and does not clearly indicate a discriminative area. Thus, I present this picture as a diagnostic example rather than as a demonstration of how the model argues. More broadly, Grad-CAM may be useful for inspection, but it should not be considered a causal explanation for the prediction.

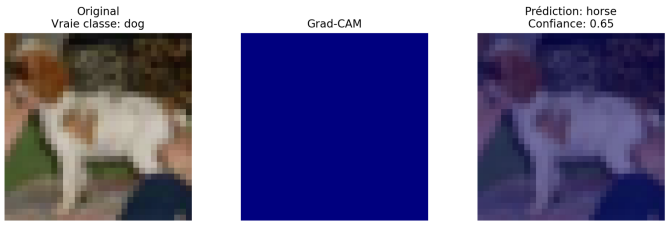

Fig. 12. Grad-CAM analysis of a misclassified CIFAR-10 picture. The real class is dog and the anticipated class is horse.

VI. DISCUSSION

The findings demonstrate a clear difference between MNIST and CIFAR-10. On MNIST, the CNN performs better than the

MLP, which implies that maintaining spa-tial information is beneficial even for a basic digit dataset. On CIFAR-10, the final CNN obtains 81.90% accuracy, but the confusion matrix also reveals that certain classes remain challenging, notably visually identical animal and vehicle classes.

The ablation research is equally essential in this publication. Dropout delivers the greatest single improvement, however integrating many regularization approaches in the complete model does not yield the best outcome. This implies that these decisions need to be verified carefully rather than presumed to assist automatically.

The data-fraction experiment reveals that additional training data increases performance, but the improvements be-come more gradual as the dataset rises. The visualization findings are valuable for examination, but they remain qualitative. In particular, Grad-CAM should be viewed as a diagnostic tool, not as verification of the model's logic.

VII. Limitations

This research has significant drawbacks. First, the trials are confined to MNIST and CIFAR-10, therefore the findings should not be applied outside these datasets. Sec-ond, CIFAR-10 employs low-resolution pictures, which makes both categorization and visual interpretation more challenging. Third, the studies are based on a restricted set of hyperparameter options and do not involve a com-prehensive multi-seed study, thus some of the observed differences may rely on the precise training setup. Fourth, VGG16, VGG19, and MobileNetV2 are not trained as complete baselines in this study and are provided just for reference. Finally, the hybrid CNN-attention model is experimental; in its present form, it should be considered as an initial test rather than as a tailored alternative to the final CNN.

VIII. Conclusion

This work provided a replicable student experimental investigation of CNNs using Keras on MNIST and CIFAR-10. The CNN improved over the MLP on MNIST, and the final CNN attained 81.90% accuracy on CIFAR-10. The trials also indicated that training choices matter: dropout offered the highest single improvement in the ablation research, and additional training data enhanced performance, albeit the gains were more slow. The hybrid CNN-attention model was effective as an exploratory test, but it did not outperform the final CNN in the given context. Finally, the visualization results helped examine the model, but they should be read cautiously. Future work might add numerous random seeds, greater augmentation, pretrained transfer learning, and further optimization of the hybrid model.

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